

HoneyBadger Lab

GLPI + Active Directory Enterprise Help Desk Implementation

Build documentation, configuration steps, validation workflow, commands, and operational testing

Environment	HoneyBadger Lab
Domain	ad.honeybadgerlab.com / HONEYBADGERLAB
Core services	Active Directory, DNS, GLPI, LDAP, Windows endpoints, file services, Wazuh, OPNsense
GLPI server	192.168.1.115 (observed in lab screenshots)
Purpose	Enterprise-style identity-integrated IT service desk and ticket workflow
Document date	October 2, 2026

Outcome: Domain users can authenticate to GLPI with Active Directory credentials. AD security-group membership is mapped to GLPI profiles such as Self-Service, Technician, and Super-Admin. A user can open an incident, a technician can receive/assign it, perform the AD remediation, document the work, and close the ticket.

1. Lab Architecture

The help-desk implementation ties the existing HoneyBadger Windows domain to GLPI through LDAP. Active Directory remains the identity source; GLPI provides service management, role mapping, ticketing, and audit history.

Component	Role in the lab
HB-WIN-DC	Windows Server domain controller; AD DS and DNS; source of users and security groups.
GLPI / Help Desk	ITSM portal at 192.168.1.115; LDAP authentication, profiles, ticket workflow.
HB-WIN11-02	Domain workstation used for end-user and technician testing.
HB-FILE-01	Domain file server; part of the broader support environment.
HB-WAZUH-01	Security monitoring / SIEM server in the broader lab.
HB-OPNsense-Firewall	Firewall/router for the lab network.
HB-KALI / Linux systems	Security and Linux administration systems used in the broader lab.

Logical flow: AD user -> LDAP authentication -> GLPI automatic authorization rule -> GLPI profile -> ticket/request workflow -> technician action -> documented resolution.

2. Server Preparation and GLPI Installation

The GLPI host was prepared as a Linux web application server. The following command set documents the standard sequence used for this build. Package names can vary slightly by Ubuntu and GLPI release.

Update the server and install the web/database/PHP stack:

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y apache2 mariadb-server
sudo apt install -y php php-mysql php-curl php-gd php-intl php-mbstring php-xml php-zip php-bz2 php-ldap
sudo systemctl enable --now apache2 mariadb
sudo systemctl status apache2
sudo systemctl status mariadb
```

Secure MariaDB and create the GLPI database:

```
sudo mysql_secure_installation
sudo mysql -u root -p

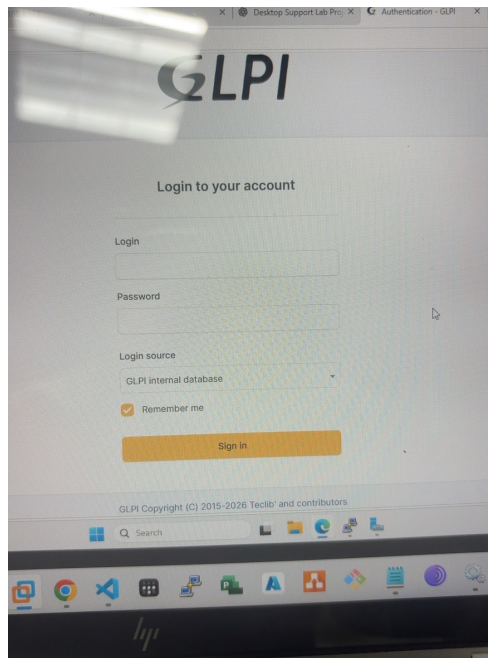
CREATE DATABASE glpidb CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER 'glpi'@'localhost' IDENTIFIED BY 'REPLACE_WITH_STRONG_PASSWORD';
GRANT ALL PRIVILEGES ON glpidb.* TO 'glpi'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

Download and place GLPI under the web root. Use the GLPI release version actually selected for the lab:

```
cd /tmp
wget https://github.com/glpi-project/glpi/releases/download/<VERSION>/glpi-<VERSION>.tgz
sudo tar -xzf glpi-<VERSION>.tgz -C /var/www/
sudo chown -R www-data:www-data /var/www/glpi
sudo find /var/www/glpi -type d -exec chmod 755 {} \;
sudo find /var/www/glpi -type f -exec chmod 644 {} \;
```

Enable Apache requirements and restart:

```
sudo a2enmod rewrite
sudo systemctl restart apache2
sudo systemctl status apache2
```



Lab evidence: GLPI installation/configuration work on the Linux server.

3. Active Directory Preparation

Active Directory security groups were created to control GLPI authorization. The important design decision is that GLPI permissions are not assigned based only on an OU; they are driven by AD group membership and GLPI authorization rules.

Security groups used in the implementation:

- **GLPI-Users** - standard users who receive the Self-Service profile.
- **GLPI-Technicians** - support personnel who receive the Technician profile.
- **GLPI-Admins** - administrators who receive the Super-Admin profile.

Example PowerShell commands for creating and validating the groups:

```
Import-Module ActiveDirectory

New-ADGroup -Name "GLPI-Users" -GroupScope Global -GroupCategory Security
New-ADGroup -Name "GLPI-Technicians" -GroupScope Global -GroupCategory Security
New-ADGroup -Name "GLPI-Admins" -GroupScope Global -GroupCategory Security

Get-ADGroup "GLPI-Users"
Get-ADGroup "GLPI-Technicians"
Get-ADGroup "GLPI-Admins"

Get-ADGroupMember "GLPI-Technicians"
Get-ADPrincipalGroupMembership "Aaron.Condit" | Select-Object Name
```

Add a user to the appropriate role group:

```
Add-ADGroupMember -Identity "GLPI-Technicians" -Members "Wines.Abigail"
Add-ADGroupMember -Identity "GLPI-Users" -Members "Aaron.Condit"

Get-ADGroupMember "GLPI-Technicians"
Get-ADGroupMember "GLPI-Users"
```

During troubleshooting, the GLPI-Technicians group was checked directly in Active Directory Users and Computers. A group with no members cannot cause the Technician authorization rule to match.

4. Configure LDAP Authentication in GLPI

In GLPI, the HoneyBadger Active Directory LDAP directory was configured under the authentication settings. The connection must be able to locate both users and groups.

Typical directory values for this lab:

Setting	Value / purpose
Directory	HoneyBadger Active Directory
Directory server	HB-WIN-DC / domain controller IP or FQDN
Base DN	DC=ad,DC=honeybadgerlab,DC=com
User login attribute	sAMAccountName
User filter	(objectClass=user) or the GLPI-recommended AD user filter
Group search type	In Groups
Group filter	(objectClass=group)
Group attribute containing users	member
User attribute containing groups	memberOf (when used by the selected search mode)

The group configuration screen in the lab showed **Search Type: In Groups**, **Filter to search in groups: (objectClass=group)**, and **Group attribute containing its users: member**. This is what allowed GLPI to evaluate the AD security groups for authorization.

5. Import and Synchronize AD Users/Groups

After LDAP connectivity was working, domain users and groups were imported/synchronized into GLPI. Authentication proves that the credentials are valid; synchronization makes the directory objects available to GLPI for role and group processing.

Validation checklist:

- LDAP connection test succeeds.
- A known AD user can be found/imported.
- GLPI-Users, GLPI-Technicians, and GLPI-Admins are visible to GLPI.
- The user's AD membership is current before testing the authorization rules.
- After changing AD membership, log out and log back in to GLPI; synchronize/import again if required.

Useful AD-side checks:

```
Get-ADUser "Aaron.Condit" -Properties MemberOf | Select-Object -ExpandProperty MemberOf
Get-ADUser "Wines.Abigail" -Properties MemberOf | Select-Object -ExpandProperty MemberOf
Get-ADGroupMember "GLPI-Users"
Get-ADGroupMember "GLPI-Technicians"
Get-ADGroupMember "GLPI-Admins"
```

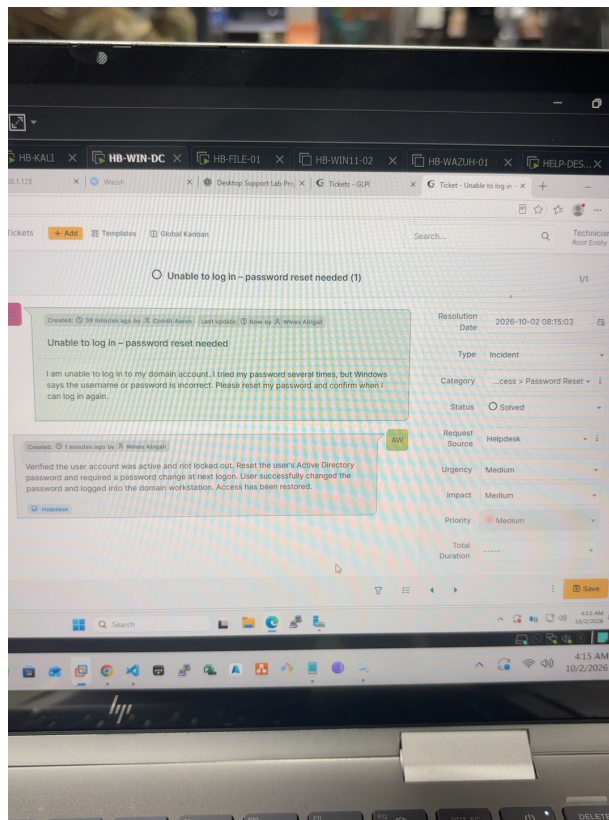
6. Build GLPI Automatic Authorization Rules

Three group-based authorization rules were created. The rules evaluate LDAP/AD group membership and assign a GLPI profile.

Position	Rule criterion	Action
1	LDAP / email authentication root rule	Assign Root Entity
2	Group is GLPI-Users	Assign profile: Self-Service
3	Group is GLPI-Technicians	Assign profile: Technician
4	Group is GLPI-Admins	Assign profile: Super-Admin

For the Technician rule, the criterion was **Global Criteria - Group is GLPI-Technicians**. The action was **Profiles -> Assign -> Technician**. The Rules Engine test returned **Validation: YES**, confirming the match.

The same model was used for GLPI-Users and GLPI-Admins. All rules should be active (green indicator).



Completed incident workflow: the technician documented the password reset and the ticket is marked Solved.

7. Troubleshooting: Why Everyone Initially Showed Self-Service

The initial symptom was that users from different AD groups still appeared in GLPI with only the Self-Service profile. The investigation isolated the issue to group membership / group processing rather than basic LDAP authentication.

Troubleshooting sequence:

- Confirmed the automatic rules existed for GLPI-Users, GLPI-Technicians, and GLPI-Admins.
- Tested the GLPI-Users rule and received Validation: YES.
- Created/validated the Technician rule with criterion Group = GLPI-Technicians and action Profile = Technician.
- Checked GLPI-Technicians in Active Directory Users and Computers and verified actual membership.
- Reviewed the LDAP group-search configuration so GLPI could read the AD **member** attribute.
- Activated all authorization rules and tested them with the GLPI Rules Engine.
- Logged in again with an account belonging to the appropriate AD group.

Result: the account mapped to the intended GLPI profile instead of remaining Self-Service.

Important lesson

LDAP authentication and authorization are separate. A user can successfully authenticate against AD while still receiving the wrong GLPI profile if group synchronization, rule criteria, rule order, or AD membership is incorrect.

8. End-User Ticket Workflow Test

A complete password-reset incident was used as the operational test.

Stage	What was tested
1. User login	Domain user authenticated to GLPI through LDAP.
2. Ticket creation	User opened a help-desk incident: Unable to log in - password reset needed.
3. Classification	Category set to Accounts & Access > Password Reset; urgency Medium.
4. Requester	Condit Aaron recorded as requester.
5. Technician access	A Technician-profile account opened the ticket.
6. Assignment	Ticket was assigned to technician Wines Abigail.
7. AD remediation	Technician reset Aaron.Condit's Active Directory password.
8. Validation	ADUC confirmed the password change; user login was verified.
9. Documentation	Technician entered the resolution notes in GLPI.
10. Closure	Ticket status changed to Solved and resolution date was recorded.

9. Password Reset Procedure Used by the Technician

The incident demonstrated a normal Tier 1 / Tier 2 desktop-support workflow. In Active Directory Users and Computers, the technician opened the user account, verified account state, reset the password, and required a password change at next logon where appropriate.

Equivalent PowerShell workflow:

```
Import-Module ActiveDirectory

# Check the account
Get-ADUser "Aaron.Condit" -Properties Enabled,LockedOut,PasswordExpired

# Unlock if required
Unlock-ADAccount -Identity "Aaron.Condit"

# Reset password interactively (preferred so the password is not stored in shell history)
$NewPassword = Read-Host "Enter temporary password" -AsSecureString
Set-ADAccountPassword -Identity "Aaron.Condit" -Reset -NewPassword $NewPassword

# Require change at next sign-in
Set-ADUser -Identity "Aaron.Condit" -ChangePasswordAtLogon $true

# Verify state
Get-ADUser "Aaron.Condit" -Properties Enabled,LockedOut,PasswordExpired,PasswordLastSet |
Select-Object SamAccountName,Enabled,LockedOut,PasswordExpired,PasswordLastSet
```

Security note: Do not put a real user's password directly into a command, ticket, screenshot, script, or documentation. Use an interactive SecureString prompt or an approved privileged-access workflow.

Resolution note entered in GLPI:

Verified the user account was active and not locked out. Reset the user's Active Directory password and required a password change at next logon. User successfully changed the password and logged into the domain workstation. Access has been restored.

10. Verification Commands and Operational Checks

```
# Windows client/domain checks
whoami
whoami /groups
ipconfig /all
nslookup ad.honeybadgerlab.com
nltest /dsgetdc:ad.honeybadgerlab.com
gpupdate /force
gpresult /r

# PowerShell AD checks
Get-ADDomain
Get-ADDomainController
Get-ADUser "Aaron.Condit" -Properties *
Get-ADGroupMember "GLPI-Technicians"

# Linux GLPI server checks
ip addr
ip route
ping -c 4 <DOMAIN_CONTROLLER_IP>
systemctl status apache2
systemctl status mariadb
php -m | grep -E 'ldap|mysqli|curl|gd|intl|mbstring|xml|zip'
sudo tail -f /var/log/apache2/error.log
```

These checks provide evidence for DNS, domain-controller discovery, group membership, web/database service health, and the PHP LDAP module required by GLPI.

11. What the Lab Now Demonstrates

- Windows Server Active Directory identity administration.
- AD security-group design for role-based access control.
- LDAP integration between a Linux-hosted enterprise application and Microsoft Active Directory.
- GLPI profile mapping through automatic authorization rules.
- Self-Service, Technician, and Super-Admin separation of duties.
- Incident intake, categorization, requester/observer/technician roles, assignment, resolution, and closure.
- A real desktop-support scenario: Active Directory password reset and account validation.
- Cross-platform troubleshooting across Windows Server, Windows clients, Linux, networking, and web services.

This is no longer just a collection of VMs. It functions as an integrated support environment in which identity, access, ticketing, endpoint administration, security monitoring, file services, and firewall infrastructure can be tested together.

12. Recommended Next Build Steps

The next logical phase is to make ticket handling more automated and measurable:

- Create ticket assignment rules by category: Password Reset -> Service Desk; Hardware -> Desktop Support; Server/AD -> Systems Administration.
- Create technician groups and assign tickets to groups instead of only individual technicians.
- Add SLAs for response and resolution times.
- Enable email notifications for ticket creation, assignment, updates, and closure.
- Deploy GLPI Inventory/Agent to domain workstations and associate devices with users.
- Integrate monitoring alerts from Wazuh into the service-management workflow.
- Build knowledge-base articles for recurring issues such as password reset, mapped drives, RDP, and domain trust.
- Create reports for open tickets, technician workload, categories, SLA compliance, and mean time to resolution.

End of build record